



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: IT25201& Foundations of Data Science using Python

Name of the Faculty member (s): Mrs. P.Ramya, AP/IT

Innovative Practice Description

Unit / Topic: Module IV / String Manipulation

Course Outcome: CO4

Topic Learning Outcome: -

Activity Chosen: Mindmap

Justification:

Using Python's string manipulation for a mind-map activity is highly effective because it transforms messy text into a structured visual format using intuitive built-in methods like `split()` and `join()` alongside advanced pattern matching via the **re module**. While standard formatting tools like **f-strings** handle the basic organization of sub-topics, integrating **Regular Expressions (regex)** allows for sophisticated data extraction, such as identifying hierarchical markers or specific keywords within unformatted notes. By leveraging functions like `re.sub()` to clean data and `re.findall()` to categorize recurring themes, you can automate the transition from a chaotic brainstorm to a precise, **dynamic system** capable of handling complex information structures with ease.

Time Allotted for the Activity: 15 Minutes

Details of the Implementation:

- The time allotted for the topic is 15 Minutes.
- After completion of Strings manipulation, the activity were conducted.
- Students are asked to draw the mind map for the given topic within 15 minutes. Each student creates their own mind map on string manipulation.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO5	PO11	PSO1
CO4	3	3	2	3	2	3

(1 – Low

2 – Moderate

3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO5	PO11	PSO1
	3	3	2	3	2	3
Justification for Correlation	Apply knowledge of computer fundamentals to solve complex problems using string algorithms.	Identify and analyze text-based problems to reach substantiated conclusions	Design system components or processes that meet specific needs (e.g., Regex for data validation).	Use modern IDEs and libraries (like Python's re or Java's StringBuffer) for efficient string handling.	Recognize the need for independent learning in the context of evolving string processing technologies (e.g., LLMs and AI).	Ability to design, implement, and evaluate software solutions using efficient data structures like Tries and Suffix Trees.

• Images / Screenshot of the practice:

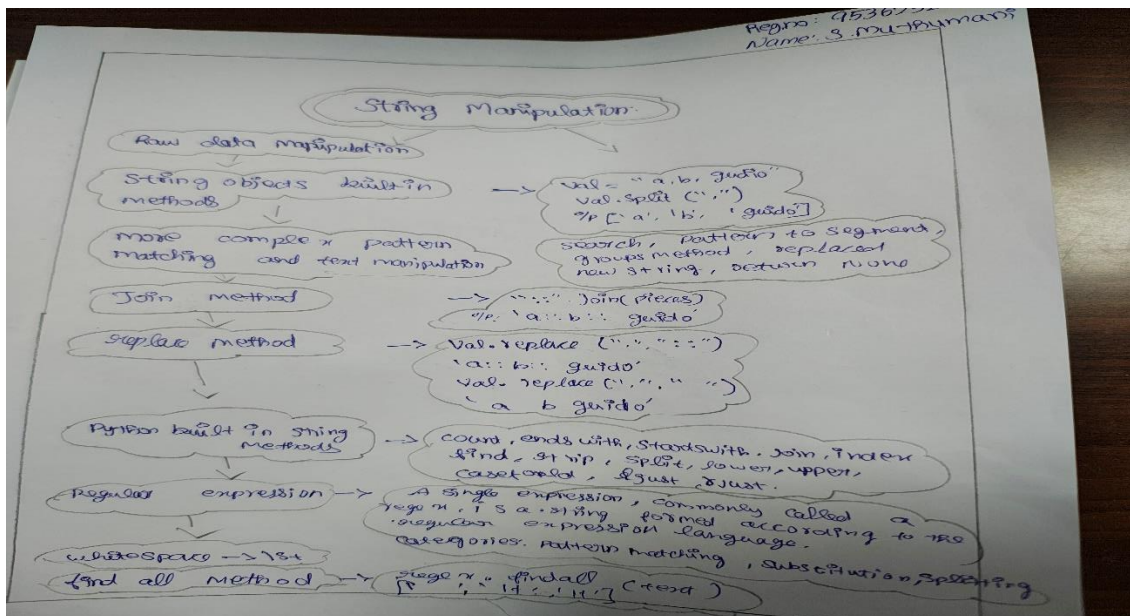


Figure 1. a Mindmap of Muthumani.M

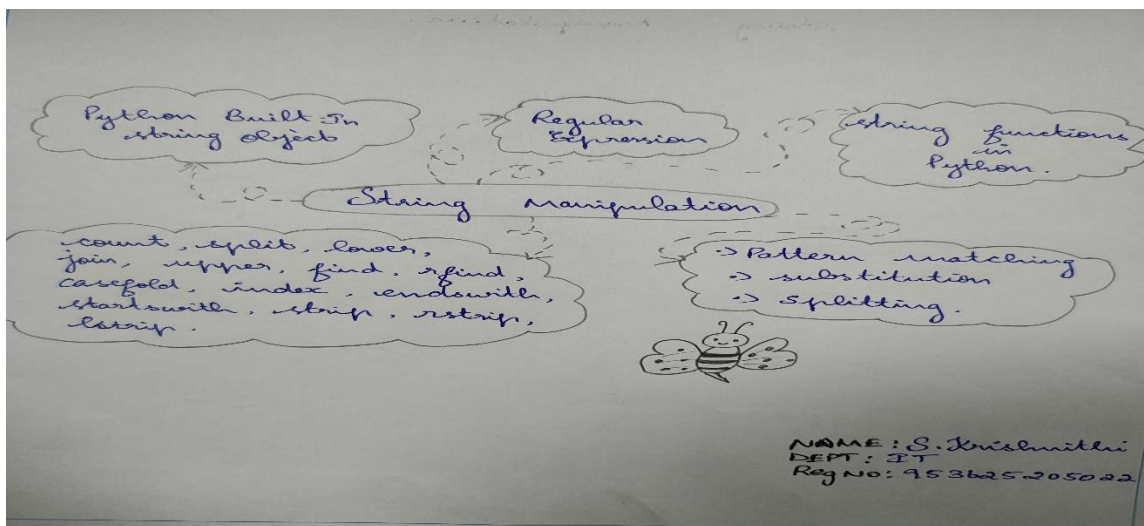


Figure 1. b Mindmap of Krishnithi. S

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students appreciated how the map grouped dozens of Built-in functions (like .find(), .replace(), .isdigit()) into logical buckets.
- Stakeholders noted that using a mind map to break down Regex syntax (quantifiers, anchors, character classes) made the "scary" symbols feel more organized.
- While students liked the map, they initially struggled to translate a branch like [a-zA-Z0-9] into a working line of code without seeing a live example.

❖ *Benefit of the practice:*

- By branching out Regular Expressions, students began to see Regex as a language of patterns rather than a random string of characters.
- Syntax Hierarchy: The map helped differentiate between Basic Methods (one-off tasks) and Regex (complex pattern matching), helping students choose the right tool for the job.
- Resource Consolidation: Students created a "single source of truth" that combined String Formatting (f-strings), Built-in Methods, and Regex patterns in one view.

❖ *Challenges faced in implementation:*

- Students struggled to show the step-by-step sequence of operations (like cleaning a string before using Regex) on a non-linear map.
- Including both Regex symbols and Built-in methods made the map too dense, leading to "information overload" and reduced readability.
- Having a function name on a branch did not guarantee students knew the correct parameters or arguments needed when actually coding.

References:

https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/KP_CS3391_Mindmap.pdf

Signature of Faculty Member

HOD